

Addressing the Challenges of Requirements Ambiguity: A Review of Empirical Literature

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Abstract—Ambiguity in natural language requirements has long been recognized as an inevitable challenge in requirements engineering (RE). Various initiatives have been taken by RE researchers to address the challenges of ambiguity. In this paper the results of a mapping study are presented that focus on the application of Natural Language Processing (NLP) techniques for addressing ambiguity in requirements. Systematic review of the literature resulted in 174 studies on the subject published during 1995 to 2015, and out of these only 28 are empirically evaluated studies that were selected. From the resulting set of papers, 81% have focused on detecting ambiguity; whereas 4% and 5% are focusing on reducing and removing ambiguity respectively. Addressing syntactic, semantic, and lexical ambiguities has attracted more attention than other types. In spite of all the research efforts, there is a lack of empirical evaluation of NLP tools and techniques for addressing ambiguity in requirements. The results have pointed out some gaps in empirical results and have raised questions the designing of an analytical framework for research in this field.

Index Terms—Ambiguity, NLP, Requirements, Systematic Mapping Study.

I. INTRODUCTION

Ambiguity is an inherent attribute of natural language (NL) and a necessary characteristic that makes NL adaptable in various contexts. It refers to the comprehension of text to hold multiple interpretations. As majority of the artefacts for requirements analysis are written in natural language [1, 2], in requirements engineering (RE), ambiguity has thus long been recognized as a challenge [3]. In RE, ambiguity is defined as “*having multiple interpretations despite the reader’s knowledge of the RE context*” [4]. Ambiguity is considered to be more intractable than other requirements’ defects such as incompleteness and that it more frequently results in misunderstandings [5]. Detecting and addressing ambiguity during RE can be more cost effective than fixing it at later stages of development. Multiple interpretations of the requirements can lead to incorrect implementation especially in case of unacknowledged ambiguity [6]. Formal and constrained languages have been proposed as an alternative to provide structure for avoiding ambiguity, however they lack the richness of NL for expressiveness of concepts [7]. Formal languages do not address the difficulties resulting from the misunderstandings stemming from expressing in NL by human judgment of the real world situations. On the contrary, these misunderstandings are documented into requirements specification and passed on to other stages of development [4].

There has been a substantial amount of research available on using Natural Language Processing (NLP) in addressing the ambiguity in natural language requirements. However, when it comes to empirical evaluation of NLP tools and techniques for addressing various aspects of ambiguities in requirements, there appears to be less interest shown from the RE research community. In this paper, results from a mapping study are presented in an attempt to aggregate the empirical work from the last two decades that address the notion of ambiguity in RE with the help of NLP tools and techniques. This mapping study has been performed by using the guidelines of systematic literature review (SLR) from Evidence Based Software Engineering (EBSE) [8]. Systematic Mapping studies (also known as Scoping Reviews) are types of secondary study recommended for providing a visual summary or a coarse-grained map of empirical research on a topic [9]. The aim of such studies is to identify gaps in a set of primary studies, and create a base for complete SLR. The results obtained provided the understanding of the state of the art in this area of research and is helpful in designing analytical framework for full systematic reviews.

II. BACKGROUND

Ambiguity is a multi-faceted and complex concept [6]. Better understanding of the nature and the source of ambiguity can help in making more informed decisions in addressing the challenges of requirements ambiguity. Ambiguity in NL arises from various sources e.g. a linguistic or syntactic problem, a conceptual problem, an invariant metaphorical interpretation etc. However identifying the source or root cause of ambiguity is a complex task [6]. Various classifications and taxonomies of ambiguity types in NL have been proposed in the literature. One notable taxonomy was by Berry [10, 11] that classified ambiguity as Lexical, Syntactic, Semantic, Pragmatic and Language errors. **Lexical ambiguity** of a word arises when a word can have several different meanings. **Syntactic ambiguity** or structural ambiguity in a sentence arises when the sentence can be parsed in more than one way, resulting in more than one grammatical structure where each provides different meanings. **Semantic ambiguity** arises in a sentence when predicate logic of a sentence can lead to multiple interpretations, although there is no lexical, syntactic, or structural ambiguity. **Language errors** ambiguity is a grammatically wrong construction that is nevertheless understood and possibly in different ways due to the error. **Pragmatic Ambiguity** is con-

cerned with the relationship between the interpretations of a sentence and its context. Pragmatic ambiguity often arises due to uncertainty of common sense knowledge and knowledge about context by humans. Some researchers have further classified the ambiguities as “*intentional and unintentional*” [7] where the writer has left ambiguity with intention or by mistake respectively, “*nocuous and innocuous*” [12] where readers can give multiple interpretations or a single interpretation to a requirements respectively, and “*nocuous ambiguity*” can be further classified as “*acknowledged and unacknowledged*” [12] whether a reader is aware of the existence of ambiguity or not. Traditionally research focus in RE is more towards avoiding or detecting the ambiguity but preserving “intentional” ambiguity has received less attention. Not all ambiguities are considered harmful. Some researchers have focused on intentionally preserving the ambiguity that can provide flexibilities during design and implementation phases [6]. It is the ‘unacknowledged’ or ‘nocuous’ ambiguities by the readers of the requirements that has high risk of leading to implementation failure [12]. Various researchers have attempted to use linguistic tools to requirements documents for avoiding, detecting and resolving ambiguity. For avoiding ambiguities, the use of formal languages, controlled natural languages, style guides, lexicons and other informal guidelines are recommended. Ambiguity can be detected in requirements by using fit criteria, test cases and inspection techniques. Some researchers used NLP tools to address challenges of ambiguity in NL requirements (e.g. QuARS [13] and LOLITA [14]).

The aim of the mapping study presented in this paper is to find all the empirical studies of the last two decades that utilizes NLP techniques, tools, methods for addressing ambiguities in natural language software and system requirements. This includes identification, extraction, removal and management of all types of ambiguities. The mapping study performed is the first step for designing an SLR on requirements ambiguity. A systematic aggregation of empirical work in this area was not available at the time of review [15]. This mapping study has identified gaps in existing empirical work on the topic and raised some important questions that provided guidance on designing analytical framework for complete SLR.

III. SYSTEMATIC MAPPING STUDY

Following the EBSE guidelines [8], a protocol was designed for search and selection of primary studies on the topic. The search string was constructed with three major terms and its variants; *natural language processing* (NLP, NL, natural language), *requirement* (requirements engineering, requirements analysis), and *ambiguity*. Six major databases were selected for search process; IEEE xplora, ACM DL, Science Direct, Springerlink, Scopus, Google Scholar. Searches were conducted in May 2015, on abstracts only and restricted to the period 1995-2015. After the results were retrieved, first the duplicates resulting from multiple search engines were discarded and then publication in different outlets based on same studies counted as single. The studies that were not in the domain of RE and were not in English or were not focusing on

requirements in English were also discarded. The remaining studies were checked as to whether they have used empirical research method (e.g. experiment, case study, survey etc.). The results were synthesised for the NLP tools and techniques employed, research methodology used, how the ambiguity was addressed, and type of ambiguity. The searches on abstracts resulted in 223 papers for the duration of 1995-2015. After discarding duplicates papers from different search engines, duplicate studies (multiple publications from the same study, only latest or journal publications were included), and irrelevant papers, 174 papers remained. From 174, only 28 have used empirical research methods to investigate ambiguity in requirements ([4, 5, 12, 16-40]). Table I provides the summary of search and selection results.

TABLE I. SUMMARY OF SEARCH AND SELECTION PROCESS

Sr#	Database	Search Results
1	IEEE xplora	28
2	ACM DL	17
3	Science Direct	6
4	Springer link	22
5	Scopus	105
6	Google Scholar	45
Total		223
Total after discarding duplicates (papers and studies) and irrelevant studies		174
Included studies		28

IV. RESULTS

Figure 1 provides an overview of the publications from 1995-2015 by indicating the number of empirical publications against the total publications on the topic.

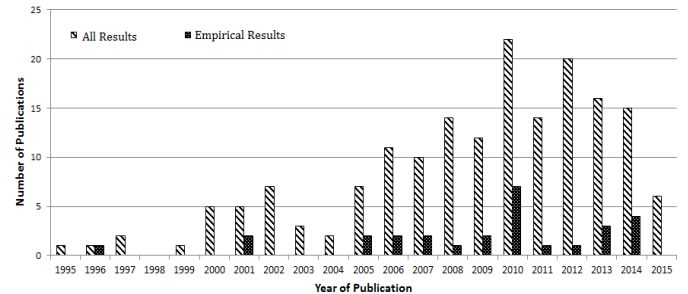


Figure 1. Ratio of empirical studies over the year

Table II gives overview of the variety of techniques that have been employed by the included 28 studies. The names of the techniques mentioned in Table II are extracted from the corresponding research papers. Out of 28 studies, 17 have used experiments as research method whereas 10 used case studies and one study was experience report. Some of the studies focused on more than one aspect of addressing ambiguity (e.g. detection, reduction and removal) and types of ambiguity. Therefore there is an overlapping in the results shown in Table IV and V. Out of 28 studies, 23 focused on detection of ambiguity, while 4 of them also addressed reducing ambiguity, and 5 also considered removal or resolution of ambiguity (Table IV). Most of the studies are focused on syntactic (16/28), semantic (5/28) and lexical (4/28) ambiguities as shown in Table V. Two studies have focused on the importance of ambiguity detection in RE [33, 40].

TABLE II. TECHNIQUES USED FOR ADDRESSING AMBIGUITY

Techniques	Reference
QuARS (Quality Analyzer for Requirements Specifications)	[18, 31, 39]
Requirements boilerplates	[34, 37, 38]
Controlled Natural Language	[20, 30]
Structural rules and templates	[23, 32]
Knowledge graph	[28, 36]
Checklist based inspection	[4, 5]
Wordnet and LOLITA	[16]
Constraint natural language	[17]
Annotating tool	[19]
Text classification and filtering	[21]
Alpino Tool	[33]
Computational Linguistics Technique: Part-of-Speech Tagging	[22]
Specialized dictionaries in NLP	[24]
Machine Learning tool	[25]
Nocuous Ambiguity Identification (NAI)	[26]
Semantic of Business Vocabulary and Rules (SBVR)	[27]
Ambiguity finding tool (AFT) - SREE	[35]
Ambiguity Taxonomy	[29]

TABLE III. RESEARCH METHODS

Research Method	Reference
Experiment	[4, 5, 12, 16-29]
Case Study	[30-38, 40]
Experience Report	[39]

TABLE IV. ADDRESSING AMBIGUITY

Addressing Ambiguity	Reference
Detecting	[4, 5, 12, 16, 18, 19, 22-26, 28-36, 38-40]
Reducing	[17, 20, 21, 27]
Removing	[5, 23, 30, 32]
Importance of ambiguity	[33, 40]

TABLE V. TYPES OF AMBIGUITY

Types of Ambiguity	Reference
Syntactic	[4, 5, 16, 18-24, 29-33, 35, 39]
Semantic	[5, 16, 19, 30, 35]
Lexical	[4, 31, 33]
Pragmatic	[28, 36]
Nocuous	[12, 25, 26]

V. DISCUSSION AND CONCLUSION

The preliminary results from this mapping study have produced the following observations:

- The RE research community has not paid enough attention to the empirical evaluation of NLP tools and techniques for addressing ambiguity in RE.
- The researchers have focused more on detection of ambiguity whereas avoiding or resolving ambiguity has been largely neglected in empirical work.
- Syntactic and semantic ambiguities are more prominent for empirical evaluation than other types.

These findings have raised some questions and concerns for further investigation by RE research community:

- The top concern is that why there is a low ratio of empirical studies in comparison to the number of publications on the topic (Fig 1). This may be due to the fact that explaining the working of a proposed technique with an example might be an easier option than designing and

executing an empirical study for verification. There are not any notable trends for replication of existing studies for verification of the proposed techniques.

- Most of the papers are focusing on detecting syntactic ambiguity which is relatively easier considering a large number of tools and techniques available in fields of linguistics and communication. It is the other types of ambiguities (e.g. semantic, pragmatic etc.) which can be difficult to detect or resolve and can cause misunderstandings and incorrect software implementation. However they are largely neglected in empirical literature which emphasizes further in depth investigation to shed light on reasons for this neglect.
- There is a lack of empirical research with specific focus on comparative analysis of the NLP tools and techniques for their effectiveness in addressing challenges of requirements ambiguity.
- There is not any research done for devising a comprehensive set of guidelines for using a specific NLP technique based on situational factors e.g. what are the contextual factors in which a certain NLP technique works better for detecting, reducing or removing a certain types of ambiguity.

VI. THREATS TO VALIDITY

The results might suffer from two major threats to validity which are considered as future work for my ongoing research. (1) There might be some studies missing from the final included sample. A secondary search strategy by using references of the included sample for identifying potential studies (snowballing) is currently in progress, and would increase the reliability of results for complete SLR. (2) The included studies at this stage are not assessed for the quality of their empirical research design. The quality assessment will be carried out in full version of SLR following EBSE guidelines [8].

VII. FUTURE WORK

The identified gaps, and questions raised from the results of this mapping study are helpful in providing guidance for designing analytical framework of a complete systematic review for our future work. The next step is to perform a meta-analysis of the resulting set of primary studies in order to find some answers to the questions and concerns that were raised from the results of mapping study. In the long term the SLR would be used to design an empirical study for evaluation of NLP tools and techniques for addressing requirements ambiguity.

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